Visible Light Positioning Using a Single LED and a Single Rotatable Photodetector

Hanyang Shi, Caiyi Wang, and Yan Zhao

***Abstract*—We investigate an indoor VLP system consisting of a single LED and a single rotatable photodetector (PD) based on the received signal strength (Liu et al., 2022). Based on the VLP system, we propose a least-square (LS) positioning algorithm. In order to analyse the influence of rod length and each rotation angle on the positioning accuracy, the Cramér-Rao lower bound (CRLB) on the positioning error of the VLP system is derived. We take advantage of the derived CRLB to choose the appropriate rod length and each rotation angle. In our VLP system, we only need to rotate the PD 4 times to achieve centimeter-level positioning accuracy. Simulation and practical experiment results show that in terms of positioning accuracy and receiver rotation times, our VLP system is significantly superior to the VLP system (Liu et al., 2022). Moreover, there is no positioning blind spots in our VLP system.**

***Index Terms*—Visible light positioning (VLP), photodetector (PD), Cramér-Rao lower bound (CRLB), least-square (LS).**

I. Introduction

WITH the development of society and economy, people's demand for indoor positioning is constantly increasing. Due to the obstruction of signal transmission, the indoor positioning accuracy of Global Positioning System (GPS) will significantly decrease [2]. Other indoor positioning technologies, such as wireless fidelity (Wi-Fi), Bluetooth Low Energy (BLE), RF identification device (RFID), Ultra-Wideband (UWB), Zigbee and Ultrasonic, have higher costs and are prone to reduced positioning accuracy due to multipath effects [3]. Compared with these positioning technologies, visible light positioning (VLP) is a promising indoor positioning technology. In addition to providing centimetre-order accurate positioning at a relatively low cost, VLP technology also has advantages such as good confidentiality, anti-electromagnetic, interference, green environmental protection, and lighting [4].

According to the required number of lamps, VLP technology can be further divided into multilamps positioning schemes and single-lamp positioning schemes. When we are located in long corridors or tunnels, there is often only one lamp installed in a large area, which leads us to adopt a single-lamp positioning schemes [5]. Moreover, when multilamps positioning schemes are adopted, multiple access techniques must be used to avoid the mutual interference among light emitting diodes (LEDs), such as Orthogonal Frequency Division Multiplexing (OFDM) [6]. In addition, some lamps being obstructed may prevent the use of the multilamps positioning schemes [7]. Therefore, the single-lamp positioning schemes is an indispensable part of indoor VLP research.

The authors in [8] proposed a novel visual odometry (VO) assisted VLP algorithm (VO-VLP), which can achieve accurate positioning using only a single luminaire at the transmitter and a single camera at the receiver. Meanwhile, in [5], the authors proposed centimeter-level 3-D mobile online VLP schemes for the UE with the multiple PDs. Additionally, in [1], the authors innovatively design a rotatable single PD receiving end, which has lower costs compared to cameras, but also avoids the decrease in positioning accuracy caused by the close distance between the multiple PDs. However, under the algorithm in [1], it needs to rotate the PD more than a hundred times to have a relatively available positioning accuracy, which is a complicated work, and the VLP system in [1] has positioning blind spots.

The main contributions of our work can be outlined as follows: 1) In order to analyse the influence of rod length and each rotation angle on the positioning accuracy of the VLP system, we derive the CRLB on the positioning error of the VLP system, which is applicable regardless of changes in rod length and each rotation angle. Based on the derived CRLB, we design the rod length and each rotation angle of the rotatable single PD receiving end. Compared to the VLP receiver in [1], our receiver can greatly reduce the number of rotations, which impove the practicality of the VLP system. 2) Based on the VLP system, we propose a least-square (LS) positioning algorithm, which can achieve centimeter-level positioning accuracy by rotating the PD 4 times. The simulations and practical experiment are designed. The results show that, in the considered indoor scenario, compared to the VLP system in [1], which requires rotating the PD 360 times, the maximum, minimum, and average positioning errors of our VLP system are significantly reduced. In addition, LS does not have positioning blind spots.

II. System Model

The indoor VLP system we are considering is illustrated in Fig. 1. Receiving end is consistent with [1]. The vertical distance from the PD to the ceiling is h, and the rod length is l. The rotation direction of the rod is shown in Fig. 1.

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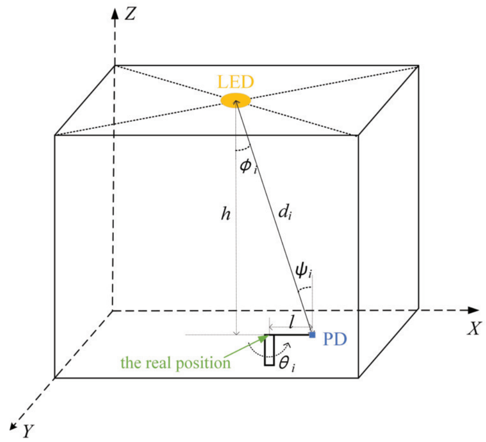

Fig. 1. Indoor VLP system with a single LED and a single rotatable PD.

We mark the three dimensional position of the LED as (X_t, Y_t, Z_t), the real position as (X_0, Y_0, Z_0), and the PD as (X_i, Y_i, Z_0). Assuming that the PD rotates from the position parallel to the positive half axis of the X-axis. We have $X_i = X_0 + l\cos\theta_i, Y_i = Y_0 - l\sin\theta_i, \theta_i = \frac{2\pi(i-1)}{N}$, $i = 1,2,...N$. Among them, N represents the number of rotations of PD, and we denote the position of PD at this time as the position after the first rotation, that is, the position of PD when $N = 1$.

Assuming that the visible light signal propagation of LED follows the Lambert radiation model, the optical channel gain for LOS transmission between LED base station and PD receiver is [9]:

$$H(0) = \begin{cases} \frac{(m+1)A_{\text{pd}}}{2\pi d^2}\cos^m(\phi)\cos(\psi)T(\psi)g(\psi), & 0 \le \psi \le \Psi \\ 0, & \psi > \Psi \end{cases} \tag{1}$$

where $m = -\frac{\log 2}{\log(\cos(\phi_{1/2}))}$, is the order of the Lambert radiation model; d is the straight-line distance between the LED base station and the PD receiver; A_{pd} is the effective photoelectric sensing area of the PD; ϕ represents the angle between the normal direction of the LED and the direction of the visible light signal emission; $\phi_{1/2}$ represents the half power angle of LED; ψ denotes the angle between the normal direction of the PD and the incident direction of the visible light signal; Ψ is the maximum incident angle of the PD; $T(\psi)$ is the optical filter gain; $g(\psi) = \frac{n^2}{\sin^2\Psi}$, is the optical concentrator gain; n is the refractive index. It should be noted that ϕ and ψ are equal when PD is placed horizontally.

After channel propagation, the visible light signal reception power of the PD receiver is:

$$P = P_{\text{led}} \cdot H(0) \tag{2}$$

where P_{led} represents the power of the LED.

In fact, PD will convert the received visible light signal into a more easily processed current signal through photoelectric conversion, and the conversion relationship can be expressed as:

$$I_{\text{r}} = R_{\text{pd}} \cdot P \tag{3}$$

where R_{pd} denotes the PD responsiveness.

In the VLP system, the total noise consists of shot noise and thermal noise, which are statistically independent and usually treated as additive Gaussian white noise [10]:

$$\sigma_{\text{n}}^2 = \sigma_{\text{shot}}^2 + \sigma_{\text{thermal}}^2 \tag{4}$$

where σ_{shot}^2, $\sigma_{\text{thermal}}^2$ can be calculated by [10].

The actual detected current signal at the receiving end is:

$$I_i = I_{\text{r}i} + n_i \tag{5}$$

where n_i denotes the noise, $n_i \sim N(0, \sigma_{\text{n}i}^2)$.

III. Positioning Method

Observing (1), (2), (3), (5), we have:

$$I_i = \frac{R_{\text{pd}}P_{\text{led}}(m+1)A_{\text{pd}}T(\psi)g(\psi)h^{m+1}}{2\pi\left[(X_0 + l\cos\theta_i - X_t)^2 + (Y_0 - l\sin\theta_i - Y_t)^2 + h^2\right]^{\frac{m+3}{2}}} + n_i \tag{6}$$

Assuming (x, y, Z_0) represents the estimated position. So, we have:

$$I_i = \frac{R_{\text{pd}}P_{\text{led}}(m+1)A_{\text{pd}}T(\psi)g(\psi)h^{m+1}}{2\pi\left[(x + l\cos\theta_i - X_t)^2 + (y - l\sin\theta_i - Y_t)^2 + h^2\right]^{\frac{m+3}{2}}} \tag{7}$$

Based on (7), we have:

$$(x + l\cos\theta_i - X_t)^2 + (y - l\sin\theta_i - Y_t)^2 = \left[\frac{2\pi I_i}{R_{\text{pd}}P_{\text{led}}(m+1)A_{\text{pd}}T(\psi)g(\psi)h^{m+1}}\right]^{\frac{-2}{m+3}} - h^2 \tag{8}$$

let $\rho_i = \left[\frac{2\pi I_i}{R_{\text{pd}}P_{\text{led}}(m+1)A_{\text{pd}}T(\psi)g(\psi)h^{m+1}}\right]^{\frac{-2}{m+3}} - h^2$.

When PD rotates N times, it will receive N different current values, N linear formulas can be obtained:

$$\begin{gathered} (x + l\cos\theta_1 - X_t)^2 + (y - l\sin\theta_1 - Y_t)^2 = \rho_1 \\ \vdots \\ (x + l\cos\theta_N - X_t)^2 + (y - l\sin\theta_N - Y_t)^2 = \rho_N \end{gathered} \tag{9}$$

Through the LS method [11], we have:

$$B = \begin{bmatrix} \rho_1 - \rho_2 + (X_t - l\cos\theta_2)^2 - (X_t - l\cos\theta_1)^2 \\ +(l\sin\theta_2 + Y_t)^2 - (l\sin\theta_1 + Y_t)^2 \\ \rho_1 - \rho_3 + (X_t - l\cos\theta_3)^2 - (X_t - l\cos\theta_1)^2 \\ +(l\sin\theta_3 + Y_t)^2 - (l\sin\theta_1 + Y_t)^2 \\ \cdots \\ \rho_1 - \rho_N + (X_t - l\cos\theta_N)^2 - (X_t - l\cos\theta_1)^2 \\ +(l\sin\theta_N + Y_t)^2 - (l\sin\theta_1 + Y_t)^2 \end{bmatrix} \tag{10}$$

$$A = \begin{bmatrix} 2(l\cos\theta_1 - l\cos\theta_2) & 2(l\sin\theta_2 - l\sin\theta_1) \\ 2(l\cos\theta_1 - l\cos\theta_3) & 2(l\sin\theta_3 - l\sin\theta_1) \\ \cdots & \cdots \\ 2(l\cos\theta_1 - l\cos\theta_N) & 2(l\sin\theta_N - l\sin\theta_1) \end{bmatrix} \tag{11}$$

$$B = A\begin{bmatrix} x \\ y \end{bmatrix} \tag{12}$$

Therefore, the estimated position of the receiver can be obtained by (13):

$$\begin{bmatrix} x \\ y \end{bmatrix} = \left(A^T A\right)^{-1} A^T B \tag{13}$$

We hope to achieve higher positioning accuracy by minimizing the number of rotations, that is, the smaller the value of N, the better, while determining a reasonable rod length l. Next, We will derive the CRLB on the positioning error of the VLP system to solve the problem.

Let $\mu = (X_0, Y_0)$. Based on (5), the probability density function (PDF) of I_i is:

$$f(I_i|\mu) = \frac{1}{\sqrt{2\pi\sigma_{\mathrm{n}i}^2}} \exp\left[-\frac{(I_i - I_{\mathrm{r}i})^2}{2\sigma_{\mathrm{n}i}^2}\right] \tag{14}$$

The current values detected by N different rotations are independent of each other. Therefore, we have the joint PDF $F(I_i|\mu)$ as:

$$F(I_i|\mu) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma_{\mathrm{n}i}^2}} \exp\left[-\frac{(I_i - I_{\mathrm{r}i})^2}{2\sigma_{\mathrm{n}i}^2}\right] \tag{15}$$

Taking the logarithm of (15), we obtain:

$$\mathrm{In}[F(I_i|\mu)] = -\sum_{i=1}^{N} \ln(\sqrt{2\pi}\sigma_{\mathrm{n}i}) - \sum_{i=1}^{N} \frac{(I_i - I_{\mathrm{r}i})^2}{2\sigma_{\mathrm{n}i}^2} \tag{16}$$

The Fisher information matrix (FIM) [12] is given by:

$$F = \begin{bmatrix} -E\left\{\frac{\partial^2 \ln[F(I_i|\mu)]}{\partial x \partial x}\right\} & -E\left\{\frac{\partial^2 \ln[F(I_i|\mu)]}{\partial x \partial y}\right\} \\ -E\left\{\frac{\partial^2 \ln[F(I_i|\mu)]}{\partial y \partial x}\right\} & -E\left\{\frac{\partial^2 \ln[F(I_i|\mu)]}{\partial y \partial y}\right\} \end{bmatrix} \tag{17}$$

Observing (16), (17), we have:

$$F = \begin{bmatrix} \sum_{i=1}^{N} \frac{1}{\sigma_{\mathrm{n}i}^2}\left[\frac{\partial I_{\mathrm{r}i}}{\partial x}\right]^2 & \sum_{i=1}^{N} \frac{1}{\sigma_{\mathrm{n}i}^2}\frac{\partial I_{\mathrm{r}i}}{\partial x}\frac{\partial I_{\mathrm{r}i}}{\partial y} \\ \sum_{i=1}^{N} \frac{1}{\sigma_{\mathrm{n}i}^2}\frac{\partial I_{\mathrm{r}i}}{\partial x}\frac{\partial I_{\mathrm{r}i}}{\partial y} & \sum_{i=1}^{N} \frac{1}{\sigma_{\mathrm{n}i}^2}\left[\frac{\partial I_{\mathrm{r}i}}{\partial y}\right]^2 \end{bmatrix} \tag{18}$$

Based on (6), we have:

$$\begin{aligned} \frac{\partial I_{\mathrm{r}i}}{\partial x} &= R_{\mathrm{pd}} P_{\mathrm{led}} \frac{(m+1)A_{\mathrm{pd}}}{2\pi} T(\psi) g(\psi) h^{m+1} \left(-\frac{m+3}{2}\right) \\ &\quad \times [(X_0 + l\cos\theta_i - X_t)^2 \\ &\quad + (Y_0 - l\sin\theta_i - Y_t)^2 + h^2]^{-\frac{m+5}{2}} \\ &\quad \times 2(X_0 + l\cos\theta_i - X_t) \end{aligned} \tag{19}$$

$$\begin{aligned} \frac{\partial I_{\mathrm{r}i}}{\partial y} &= R_{\mathrm{pd}} P_{\mathrm{led}} \frac{(m+1)A_{\mathrm{pd}}}{2\pi} T(\psi) g(\psi) h^{m+1} \left(-\frac{m+3}{2}\right) \\ &\quad \times [(X_0 + l\cos\theta_i - X_t)^2 \\ &\quad + (Y_0 - l\sin\theta_i - Y_t)^2 + h^2]^{-\frac{m+5}{2}} \\ &\quad \times 2(Y_0 - l\sin\theta_i - Y_t) \end{aligned} \tag{20}$$

Since localization can be considered as an unbiased estimation problem, we can derive the lower bound of the expected estimation error $\mathbb{E}[(X_0 - x)^2 + (Y_0 - y)^2]$:

$$\mathbb{E}[(X_0 - x)^2 + (Y_0 - y)^2] \geq \mathrm{trace}(F^{-1}) = \mathrm{CRLB} \tag{21}$$

When the rod length l and rotation angle θ_i are determined, the value of CRLB is uniquely determined. From equation (21), it can be seen that the smaller the value of CRLB, the higher the positioning accuracy. We set appropriate rod length and each rotation angle by comparing the CRLB values at the same position under different conditions.

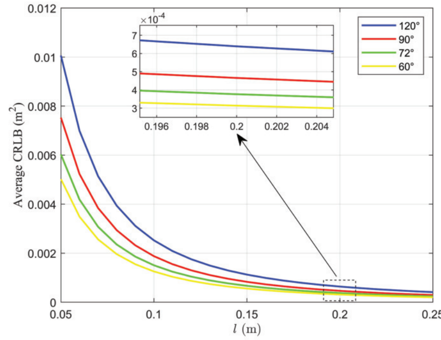

Fig. 2. Average CRLB at different rod length and each rotation angle.

We define the error function as:

$$d_{\mathrm{error}} = \sqrt{(X_0 - x)^2 + (Y_0 - y)^2} \tag{22}$$

IV. Simulation and Experimental Results

A computer simulation was conducted on the MATLAB platform in an indoor scene measuring $5\,\mathrm{m} \times 5\,\mathrm{m} \times 4\,\mathrm{m}$. The coordinates of the LED are $(2.5, 2.5, 4)$. $P_{\mathrm{led}} = 20\mathrm{W}$, $\phi_{1/2} = 60°$, $n = 1.5$, $\Psi = 80°$, $T(\psi) = 1$, $R_{\mathrm{pd}} = 0.54\mathrm{A/W}$, $A_{\mathrm{pd}} = 1\mathrm{cm}^2$. The remaining parameter values are consistent with [10]. Our point selection rule in the plane is as follows: the range of X and Y is from 0.2 to 4.8, with a step size of 0.2, and there are a total of 576 uniformly distributed points, all of which are within the field of view of the PD.

Fig. 2 discusses the influence of rod length and each rotation angle on the average CRLB (mean of CRLB at 576 points) when $h = 3\mathrm{m}$, $\sigma_{\mathrm{n}}^2 = 4.2291 \times 10^{-15}\,\mathrm{A}^2$ (mean of σ_{n}^2 at 576 points). Considering practical applications, the rod length range we are discussing is from 5cm to 25cm, with each rotation angle of 120 degrees, 90 degrees, 72 degrees, and 60 degrees, respectively. The simulation results indicate that an increase in rod length and a decrease in each rotation angle will lead to an improvement in positioning accuracy. This is because as the length of the rod increases, there is a greater difference in the power values of the optical signals collected at different angles; As each rotation angle decreases, the number of rotations increases, and the more information is collected. Based on the simulation results, we set the rod length to 20cm and rotate it 90 degrees each time.

Fig. 3(a) shows the distribution of positioning errors through LS when $h = 3\mathrm{m}$. The maximum positioning error is 0.0843m, the minimum positioning error is 0.0080m, and the average positioning error is 0.0252m. The positioning errors in the four corners are significantly higher than those in the central area. This is because the LED is installed at the center of the roof, and the closer the positioning point is to the center, the higher the received light power and the better the positioning accuracy. Fig. 3(b) shows the distribution of positioning errors through the direct positioning algorithm in [1] when $h = 3\mathrm{m}$ and rotates 360 times by 1 degree each time. The maximum positioning error is 0.2014m, the minimum positioning error

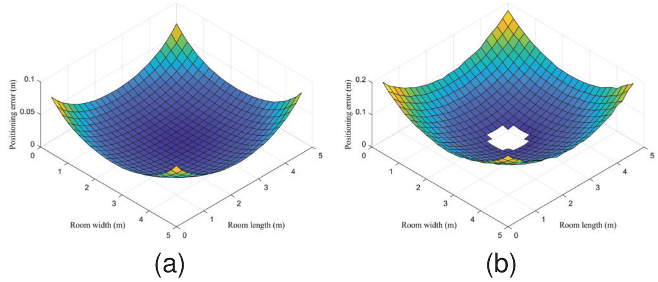

Fig. 3. Distribution of positioning errors.

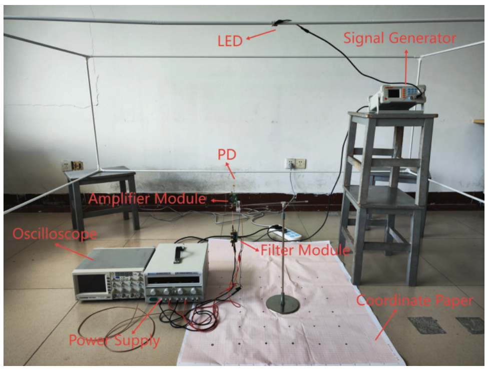

Fig. 4. Experimental setup.

is 0.0244m, and the average positioning error is 0.0863m, all of which are much larger than the positioning errors shown in Fig. 3(a). The white area in the middle represents the blind spot for positioning. When simulating two algorithms, the length of the rod and the initial rotation position are the same. Fig. 3 indicates that our algorithm achieves much higher positioning accuracy after 4 rotations than the algorithm in [1] after 360 rotations. In the algorithm in [1], the author needs to accurately find the closest and farthest positions of the PD from the LED, that is, the positions with the highest and lowest light power, and records the corresponding rotation angles of the two positions. However, due to the influence of noise and each rotation angle, it is difficult to accurately find these two positions, which makes the positioning accuracy of the algorithm itself low. When the positioning point is located directly below the LED, due to the noise interference, the measured current at the receiving end may be lower than the real value. If the reduction is too much, it will cause d_i to be less than h, and the algorithm in [1] cannot be used.

Fig. 4 and Fig. 5 show the experimental setup and results, respectively. In the experiment, $h = 0.59$m, $P_{\text{led}} = 5$W, $\phi_{1/2} = 60^\circ$, $R_{\text{pd}} = 0.39$A/W, $A_{\text{pd}} = 150\text{mm}^2$. The experiment was conducted in an enclosed room. We use a signal generator to generate a sine signal with a frequency of 500 kHz. The frequency of sine signal is set to be 500 kHz for good separation with the near-DC component including DC bias and ambient light. Positioning operations were performed on 36 test points within the designated area. By repeating the same experiment for the 36 test points in the positioning area, the estimated position coordinates and positioning errors corresponding to the 36 test points were obtained. A (41.06, 49.11) is the point with the smallest error. B (13.12, 14.29) is the point with the largest error. The average positioning error is 4.9613cm.

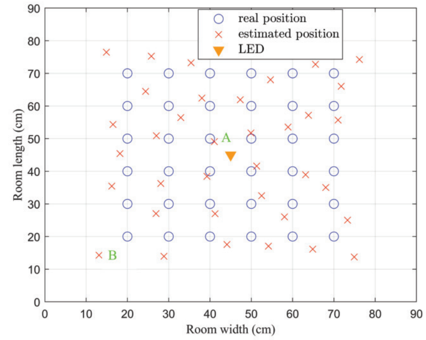

Fig. 5. Experimental results.

V. Conclusion

In this letter, we investigate an indoor VLP system consisting of a single LED and a single rotatable PD based on the received signal strength. We develop a LS positioning algorithm and derive the CRLB on the positioning error for the VLP system. Based on the VLP system, we only need to rotate the PD 4 times to achieve centimeter-level positioning accuracy. Simulation and experimental results show that compared to the VLP system in [1], our VLP system is simpler, with higher positioning accuracy and no positioning blind spots. Future works include studying the influence of the ambient light on the VLP system.

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